

Understanding Logs

A **log** is a **record of events or activities** that occur on a computer system, network, application, or security device.

It helps security professionals understand:

- **What happened?**
- **When did it happen?**
- **Where did it happen?**
- **Who performed the action?**
- **Was the action successful or did it fail?**

A log does **not always directly tell you why or how** an incident happened. Instead, it provides evidence that security professionals analyze to determine the cause.

Common Information in a Log

A log entry usually contains:

- Timestamp (Date & Time)
- Username or Account
- Source IP Address
- Destination IP Address (if applicable)
- Hostname or Device Name
- Event ID
- Action Performed
- Status (Success or Failure)
- Event Description

Types of Logs

There are many different types of logs, including:

- System Logs
- Security Logs
- Application Logs
- Network Logs
- Firewall Logs
- Authentication Logs
- Audit Logs
- Database Logs
- Web Server Logs
- DNS Logs
- Email Logs
- Cloud Logs
- IDS/IPS Logs

Each log records different types of events.

Logs in Incident Response

From a cybersecurity perspective, security analysts should **never rely on a single log source**.

Instead, they collect and correlate logs from multiple systems using a **SIEM (Security Information and Event Management)** platform.

During an incident, analysts use logs to answer questions such as:

- What happened?
- When did it happen?
- Where did it happen?
- Which systems were affected?
- Which user or account performed the action?

- Was the action successful or blocked?
- How did the attacker move through the environment?
- What is the impact of the incident?

After analyzing the logs, the security team can:

1. Detect the incident.
2. Investigate the root cause.
3. Identify the affected systems.
4. Contain the attack.
5. Eradicate the threat.
6. Recover the affected systems.
7. Document lessons learned to improve future security.

Simple Understanding

Logs are the digital evidence of everything that happens inside an IT environment.

Security professionals analyze multiple log sources to reconstruct the timeline of an incident, determine what occurred, identify who was involved, understand how the attack progressed, and respond effectively.

One important correction

"log explain why happened and how happened"

This is **not always true**.

A log records **events**, but it may not explicitly explain **why** something happened.

For example:

10:30:15

User: admin

Status: Failed Login

IP: 203.0.113.15

The log tells you:

-  What happened? → Failed login
-  When? → 10:30:15
-  Who? → admin account
-  From where? → 203.0.113.15

But it **doesn't tell you why** the login failed. It could be because:

- The user entered the wrong password.
- The account was locked.
- An attacker was performing a brute-force attack.
- MFA failed.

To determine the **why** and **how**, analysts correlate this log with **other logs** (authentication logs, firewall logs, Active Directory logs, EDR logs, etc.). This process is called **log correlation**, and it's one of the core skills of a SOC Analyst.

SIEM (Security Information and Event Management)

What is SIEM?

SIEM stands for **Security Information and Event Management**.

It is a centralized security platform that **collects, stores, normalizes, correlates, analyzes, and monitors logs** from multiple devices, systems, and applications to detect cyber threats.

Simple Definition

A SIEM collects logs from different sources, analyzes them together, detects suspicious activities, generates alerts, and helps security analysts investigate security incidents.

Think of a SIEM as the **brain of a Security Operations Center (SOC).**

Why Do We Need SIEM?

Every organization has many devices generating logs every second.

Examples:

- Windows Computers
- Linux Servers
- Firewalls
- Routers
- Switches
- Databases
- Cloud Services
- Applications
- VPN
- Active Directory
- IDS/IPS

Without SIEM, analysts would need to manually check every device separately.

SIEM centralizes all logs into one platform, making monitoring and investigation much faster.

SIEM Functions

A SIEM performs six major functions.

1. Log Collection

Collects logs from multiple devices and applications.

Examples:

- Windows Event Logs
 - Linux Logs
 - Firewall Logs
 - Network Devices
 - Applications
 - Cloud Services
 - Databases
-

2. Log Normalization

Different devices produce logs in different formats.

The SIEM converts them into a common format so they can be analyzed together.

3. Log Correlation

One of the most important SIEM features.

Instead of analyzing one log, the SIEM connects related events from different systems.

Example:

Firewall Log

↓

VPN Login

↓

Failed Login Attempts

↓

PowerShell Execution

↓

Administrator Account Created

↓

Possible Account Compromise

4. Threat Detection

The SIEM detects suspicious activities using rules, analytics, and threat intelligence.

Examples:

- Brute Force Attack
 - Malware Activity
 - Privilege Escalation
 - Suspicious PowerShell Commands
-

5. Alerting

When suspicious activity is detected, the SIEM generates alerts.

These alerts are investigated by SOC Analysts.

6. Investigation and Reporting

Security analysts search logs to answer:

- What happened?
- When did it happen?
- Which user was involved?

- Which IP Address?
 - Which system was affected?
 - What actions occurred?
-

SIEM Workflow

Logs Generated



Collected by SIEM



Normalized



Correlated



Threat Detection



Alert Generated



SOC Analyst Investigation



Incident Response

Popular SIEM Tools

Some of the most widely used SIEM platforms include:

- Splunk Enterprise Security

- Microsoft Sentinel
 - IBM QRadar
 - ArcSight
 - Elastic Security
 - Wazuh
 - Google Security Operations
-

Splunk

What is Splunk?

Splunk is a **commercial (paid) SIEM platform** widely used by enterprises for log management, threat detection, and security monitoring.

Although Splunk offers free and trial editions, most production environments use licensed enterprise versions.

Features

- Centralized Log Collection
 - Powerful Dashboards
 - Threat Detection
 - Alerting
 - Reporting
 - Compliance Monitoring
 - Powerful Search Language (SPL)
-

Splunk Search Language (SPL)

Splunk uses **Search Processing Language (SPL)** to search and analyze logs.

Example:

index=windows EventCode=4625

This searches for Windows failed login events.

Advantages

- Industry Standard
 - Excellent Search Capabilities
 - Large Enterprise Support
 - Strong Dashboards
 - High Scalability
-

Disadvantages

- Expensive
 - Commercial Licensing
 - Can require significant resources
-

Wazuh

What is Wazuh?

Wazuh is a **free and open-source SIEM and XDR platform**.

Besides log management, it provides endpoint monitoring and security monitoring capabilities.

Features

- Log Collection
- Endpoint Monitoring

- File Integrity Monitoring (FIM)
 - Vulnerability Detection
 - Threat Detection
 - Security Monitoring
 - Compliance Monitoring
-

Architecture

Endpoint Agent



Wazuh Manager



Indexer



Dashboard



SOC Analyst

Agents installed on endpoints collect security events and send them to the Wazuh Manager.

Advantages

- Completely Free
- Open Source
- Easy to build a home lab
- Built-in Endpoint Security Features
- Excellent for Learning

Disadvantages

- Requires self-management
 - Smaller enterprise ecosystem than Splunk
 - Less mature search capabilities compared to SPL
-

Splunk vs Wazuh

Feature	Splunk	Wazuh
License	Commercial (Paid)	Free & Open Source
Type	SIEM	SIEM + XDR
Best For	Large Enterprises	Learning, Home Labs, SMEs
Endpoint Monitoring	Requires integrations	Built-in
File Integrity Monitoring	Available through integrations	Built-in
Search Language	SPL	OpenSearch/Elastic Queries
Dashboards	Excellent	Good
Threat Detection	Excellent	Very Good
Community	Large Enterprise Community	Large Open Source Community

Which One Should You Learn?

For Beginners

Start with **Wazuh** because:

- It is free.

- You can install it in your own lab.
 - It teaches SIEM fundamentals.
 - It includes endpoint monitoring and threat detection.
-

For Professional Jobs

Learn **Splunk** because:

- It is one of the most widely used enterprise SIEM platforms.
 - Many SOC Analyst job descriptions require Splunk experience.
 - Learning SPL is valuable for interviews and real-world environments.
-

Recommended Learning Path

1. Windows Event Logs
 2. Linux Logs
 3. Sysmon
 4. Wazuh
 5. Splunk (SPL)
 6. Microsoft Sentinel
-

XDR, EDR & File Integrity Monitoring (FIM)

EDR (Endpoint Detection and Response)

Definition

EDR stands for **Endpoint Detection and Response**.

It is a security solution that **monitors, detects, investigates, and responds to threats on endpoint devices**.

What is an Endpoint?

An endpoint is any device connected to a network, such as:

- Desktop
- Laptop
- Windows Server
- Linux Server
- Virtual Machine

Purpose

EDR protects individual endpoints by monitoring:

- Running processes
- User activity
- Malware
- PowerShell commands
- Registry changes
- File modifications

Simple Definition

EDR protects endpoint devices by detecting suspicious activities and helping security teams investigate and respond to attacks.

XDR (Extended Detection and Response)

Definition

XDR stands for Extended Detection and Response.

It is an advanced security platform that collects, correlates, detects, investigates, and responds to threats across multiple security sources from one centralized platform.

Unlike EDR, XDR is not limited to endpoints.

It combines data from:

- Endpoints
- Email
- Firewall
- Network
- Cloud
- Identity Management
- SIEM

Purpose

XDR provides a complete view of an attack by correlating events from different security tools.

Simple Definition

XDR extends EDR by combining security data from multiple sources to detect, investigate, and respond to cyber attacks more effectively.

File Integrity Monitoring (FIM)

Definition

File Integrity Monitoring (FIM) continuously monitors important files and directories to detect unauthorized changes.

It alerts security teams whenever a file is:

- Created
- Modified
- Deleted
- Renamed
- Permission Changed

- Owner Changed

FIM uses **cryptographic hashes (such as SHA-256)** to determine whether a file has changed.

Purpose

Protect critical system and application files from unauthorized modifications.

Simple Definition

FIM monitors important files and immediately alerts security teams if they are changed, deleted, or tampered with.

EDR vs XDR

EDR	XDR
Protects endpoint devices	Protects the entire security environment
Focuses on one endpoint	Correlates multiple security sources
Detects endpoint threats	Detects attacks across endpoints, networks, email, cloud, and identity systems
Limited scope	Broader visibility and automated response

One-Line Summary

- **EDR:** Detects and responds to threats on endpoint devices.
 - **XDR:** Detects and responds to threats across multiple security systems from one platform.
 - **FIM:** Monitors critical files and alerts when they are modified, deleted, or tampered with.
-

